Title: TOWARD ROBUST DEEP LEARNING: EVALUATING UNCERTAINTY IN TRANSFORMER-BASED ARCHITECTURES FOR SCIENTIFIC DATA

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Abstract:
Deep Learning (DL) models, specifically Convolutional Neural Networks (CNNs) and Transformers, have demonstrated significant efficacy across various scientific domains. However, their integration into critical research is often constrained by the "black-box" nature of these models and their tendency to produce overconfident predictions when processing noisy or out-of-distribution datasets. This study investigates the implementation of robust DL architectures, focusing on the fine-tuning of Transformer models to optimize predictive accuracy while ensuring rigorous model calibration. We analyze the propagation of uncertainty through deep layers and evaluate the impact of targeted feature selection on overall model stability. By comparing standard architectures with uncertainty-aware configurations, this research establishes a methodology for developing more reliable and interpretable computational pipelines. Preliminary results indicate that incorporating Bayesian-inspired uncertainty estimation significantly enhances the robustness of Transformer models in high-dimensional scientific environments, offering a scalable path toward more sustainable and portable machine learning solutions.

References:
[1] Y. Gal and Z. Ghahramani, Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning, Proc. Int. Conf. Mach. Learn. 33, 1050 (2016).
[2] A. Vaswani et al., Attention is All You Need, Adv. Neural Inf. Process. Syst. 30, 5998 (2017).